

# Muhammad Ubaidullah

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## SUMMARY

Computer Science graduate with internship experience in Data Engineering and hands-on expertise in backend development, computer vision, blockchain, and workflow automation. Experienced in building scalable Django applications, PostgreSQL databases, and AI-powered systems. Seeking an Associate Software Engineer or Backend Engineer role where I can contribute to building reliable and high-performance software solutions.

## EDUCATION

**Information Technology University**  
*Bachelor of Science in Computer Science*

Lahore, Punjab  
*Aug 2022 – July 2026*

**Punjab Group of College**  
*FSC Pre-Engineering*

Lahore, Punjab  
*October 2020 – June 2022*

## EXPERIENCE

**Data Engineer Intern**  
*Axiom World*

June 2025 – August 2025

- Automated data collection workflows using n8n and Selenium, reducing repetitive manual processing across multiple datasets.
- Managed structured datasets across raw, silver, gold, and production layers while ensuring consistent data quality.
- Designed ETL pipelines and data schemas supporting efficient transformation and classification of public and restricted datasets.
- Integrated machine learning outputs with Django-based backend services through APIs, enabling real-time data availability.
- Optimized backend APIs and PostgreSQL operations to improve application performance and data flow
- Collaborated with cross-functional teams to deploy data-driven solutions using Django, databases, and automation tools.

## PROJECTS

**Face Recognition System for Low-Resource Devices.**

**A Comparative Analysis of State-of-the-Art Face Recognition Models**

*Python, Django, OpenCV, InsightFace, ArcFace, RetinaFace, ONNX Runtime, PostgreSQL*

- Evaluated five state-of-the-art face recognition models on the SCFace surveillance dataset, selecting ArcFace for its superior accuracy and CPU deployment efficiency.
- Built a desktop and web-based attendance management system with real-time face recognition, automated attendance, and centralized administration.
- Designed an optimized face recognition pipeline using RetinaFace, ArcFace embeddings, cosine similarity, and ONNX Runtime for efficient inference on low-resource devices.
- Created a custom low-resource dataset of 30 subjects and benchmarked model accuracy, latency, memory usage, and model size under challenging real-world conditions.
- Integrated PostgreSQL, email alerts, NTFY mobile notifications, and DroidCam-based live monitoring to automate attendance tracking and unknown-face reporting.

**Blockchain-Based Donation System with AML Monitoring**

*Django, Solidity, Ethereum, Web3.py, Hardhat, MetaMask*

- Built a blockchain-powered donation platform enabling transparent Ethereum-based donations with immutable transaction records.
- Engineered Solidity smart contracts using Hardhat to securely manage donations and charity interactions.
- Connected Django backend with Ethereum using Web3.py to retrieve and process on-chain transaction data.
- Integrated MetaMask for wallet-based authentication and transaction signing, allowing donors to securely initiate and authorize cryptocurrency donations directly from their Ethereum accounts.
- Designed an AML monitoring module allowing administrators to track, review, and flag suspicious blockchain transactions.

- Developed role-based functionality for donors, charities, and administrative authorities, with centralized monitoring and management of users, donations, charities, and flagged blockchain transactions.

#### **C++ Compiler From Scratch** | *C++*

- Built a toy C++ compiler from scratch in C++ to explore core compiler design concepts.
- Implemented lexical analysis to tokenize input source code.
- Developed a recursive-descent parser for syntax analysis and basic semantic checks.
- Handled error detection and parsing of grammar defined in a custom specification.
- Explored the foundations of compiler pipelines, including future support areas such as code generation.

#### **Speech Transcriber & Translator** | *Python, OpenAI Whisper, Google Translate API, Tkinter*

- Built a desktop application that records live audio or accepts uploaded audio files for automatic speech-to-text transcription using OpenAI's Whisper model.
- Integrated the Google Translate API to translate transcribed text into multiple languages with a simple and intuitive user interface.
- Implemented features to display, copy, and download translated text, providing a complete end-to-end speech processing workflow.
- Designed a lightweight desktop application supporting offline transcription with online translation, enabling efficient multilingual communication.

#### **Airbnb Clone** | *MERN Stack*

- Developed a full-stack Airbnb-inspired web application using React, Node.js, and MongoDB.
- Implemented JWT-based authentication, role-based access control, and protected admin routes.
- Built features for property listing, booking management, and host-specific CRUD operations.

#### **College Management System** | *Csharp, ASP.NET, MySQL, HTML, CSS, Bootstrap*

- Developed a full-stack web application using the .NET framework
- Designed a MySQL database to store records for college
- Implemented role-based access (Admin and Instructor) for secure and efficient system usage

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## TECHNICAL SKILLS

**Programming Languages:** C/C++, Python, C#, JavaScript, HTML/CSS, XML

**Frameworks:** .NET, React.js, Node.js, Django

**Databases:** MongoDB, MySQL, PostgreSQL, NocoDB

**Libraries:** OpenCV, InsightFace, Pandas, NumPy, Matplotlib, PyTorch, Tkinter, SFML, Selenium

**Tools & Environments:** Git, VS Code, Visual Studio Community, Docker, Jupyter, Linux, n8n